

Brainstorming & Idea Prioritization

SmartBridge AI/ML Internship — HDI Prediction System

1. Objective

To brainstorm possible AI/ML project ideas within the team, evaluate each idea against feasibility, impact, and available resources, and select the most suitable idea to build as our SmartBridge internship project.

2. Brainstorming Session Summary

A team brainstorming session was conducted where every member contributed at least one project idea related to real-world data and machine learning. The ideas were discussed for relevance, data availability, and social impact.

3. List of Ideas Generated

- 1 Human Development Index (HDI) Prediction System using country-wise socio-economic indicators
- 2 Student Performance Prediction using academic and demographic data
- 3 House Price Prediction based on location and property features
- 4 Employee Attrition Prediction for HR analytics
- 5 Crop Yield Prediction using agricultural and weather data

4. Idea Prioritization Matrix

Idea	Feasibility	Impact	Data Availability	Priority
HDI Prediction System	High	High	High	1 (Selected)
Student Performance Prediction	High	Medium	Medium	2
House Price Prediction	Medium	Medium	High	3
Employee Attrition Prediction	Medium	Medium	Medium	4
Crop Yield Prediction	Low	High	Low	5

5. Selected Idea & Justification

The team selected the **Human Development Index (HDI) Prediction System** as the final project idea. This idea scored highest across feasibility, real-world impact, and data availability, since a well-structured global HDI dataset (covering around 195 countries with 2021 socio-economic indicators) was readily available. Predicting HDI also connects machine learning with meaningful social and economic themes such as education, health, and income, making it a strong and demonstrable capstone project for the internship.

6. Next Steps

- Define the problem statement clearly.
- Collect and clean the HDI dataset.
- Train a Linear Regression model to predict HDI values.
- Build a Flask web application for real-time predictions.
- Deploy the application on Render.com for public access.